
BLEI-E Index

Bledsoe Enlightenment Performance Research Index

Quantitative Financial Product Researching

Author: Dashawn Ramel Bledsoe | **Position:** Blindside Flanker | **Club:** New York Rugby Club (NYRC)

League: Division I USA Rugby — Empire GU | **GitHub:** drb-apes | **Status:** Active Research Development

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Research Status: Active **Version:** 1.0 **License:** Proprietary **Index Type:** Performance Bond **League:** Division I USA Rugby **Platform:** BiomechFlow Capsule

Proprietary solo-athlete quantitative research index powered by BiomechFlow Capsule biomechanical data captured during live training sessions and competitive matches. Authored and owned solely by Dashawn Ramel Bledsoe as an individual intellectual property asset — no corporate entity required.

About This Project

The **BLEI-E Index** (Bledsoe Enlightenment Performance Research Index) is a proprietary, session-logged quantitative performance research index built on **BiomechFlow Capsule** biomechanical data captured during live training sessions and competitive matches at the **New York Rugby Club (NYRC)**, Manhattan, NY. It is authored and owned solely by **Dashawn Ramel Bledsoe** as an individual intellectual property asset — no corporate entity required.

The index translates six sensor-driven capsule zone outputs into a weighted composite performance score — the **BLEI-E Score** — updated per session and per match. The index is designed for licensing to quantitative research firms, sports analytics companies, and financial institutions as a novel, underrepresented data asset in the US rugby market.

The **Dashawn Bledsoe Bond** is the financial instrument built on this index as its underlying performance metric — analogous to the Jalen Brunson Bond Index model, but anchored in verifiable, auditable biomechanical data rather than market sentiment. This repository serves as the authoritative public-facing project proposal, methodology reference, and licensing documentation hub for the BLEI-E Index ecosystem.

Why US Rugby? Why Now?

US rugby is one of the most data-sparse professional athletic markets in North America. Division I club rugby generates minimal indexed biomechanical data, creating a first-mover advantage for individual athletes willing to instrument their own performance. BLEI-E is designed to fill that gap with a rigorous, auditable, and licensable dataset.

Repository Topics / Tags

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NIL

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USA Rugby

NYRC

Player Profile

Field	Detail
Player Name	Dashawn Ramel Bledsoe
Alias	"Matrix"
Position	Blindside Flanker
Team	New York Rugby Club (NYRC)
League	Division I USA Rugby — Empire GU
Location	Manhattan, New York City
Academic Credential	Columbia University — Sports Management / Event Management
Index Role	Sole Author and Licensor

The Dashawn Bledsoe Bond

The **Dashawn Bledsoe Bond** is a personal performance bond instrument whose underlying value is indexed directly to the **BLEI-E Composite Score**. It is the first rugby-data-driven individual athlete performance bond in the US market — a novel financial instrument that treats a Division I rugby player's biomechanically verified performance trajectory as a structured, tradeable research asset.

Unlike traditional athlete endorsement deals or generic NIL agreements, the Dashawn Bledsoe Bond is engineered from the ground up with quantifiable, session-level data as its foundational layer. Every coupon-equivalent payment, every maturity trigger, and every risk scenario is tied to an observable, auditable BLEI-E signal — not sentiment, speculation, or subjective scouting opinion.

Bond Mechanics

Component	Detail
Underlying Asset	BLEI-E Composite Score (0-100)
Par Value	\$10,000 per unit (research denomination)
Coupon Equivalent	Data licensing fees + NIL sponsor activation income
Coupon Range	4-8% annually, scaling with dataset maturity and licensing breadth
Maturity Trigger 1	Professional contract signing — matures at 110% par
Maturity Trigger 2	USA Rugby international selection — matures at 125% par
Maturity Trigger 3	BLEI-E cited in published research / licensed to \$100M+ AUM fund — matures at 105% par
Default Scenario	Rolls into perpetual research royalty instrument
Legal Pathway	SEC Reg D 506(b) private placement (Year 2+)

Legal Notice

All bond framework references herein are for research and conceptual development purposes only. No formal bond issuance is active. Any commercial issuance requires full SEC Reg D 506(b) compliance and engagement of independent licensed securities counsel. See Disclaimer section.

BLEI-E Signal Architecture — The 6-Factor Model

The BLEI-E Score is a weighted composite of six capsule zone outputs, each mapped to a specific BiomechFlow Capsule zone on the player's body. Each factor is assigned a weight reflecting its contribution to overall performance valuation. Trigger thresholds define when a factor generates an actionable signal within the 24-Method HEDGE Module.

Factor	Capsule Zone	Weight	Metric Captured	Signal Trigger
Contact Power Index	Top Arc	25%	Tackle Force (N) + Shoulder Contact Angle	Stress spike >+20% vs. session baseline
Drive & Fatigue Load	Bottom Arc	20%	Sled Push Velocity + HRV Decay Rate	Fatigue color shift to red zone
Recovery Velocity	Center Line	15%	HRV Recovery Slope + Inter-effort Rest Duration	Recovery slope divergence from 7-day mean
Injury Risk Premium	Right Bar	15%	Joint Divergence Score + Impact Vector Alignment	Risk elevation >+15% above rolling average
NIL Momentum Score	Glow Ring	15%	Social Engagement Rate + Highlight Clip	Momentum surge >+25% week-over-week

Factor	Capsule Zone	Weight	Metric Captured	Signal Trigger
			Frequency	
Substitution Probability	Left Bar	10%	Substitution Probability Model Output	Probability >60% in live game mode

BLEI-E Composite Formula

BLEI-E Score = (Top Arc × 0.25) [Contact Power Index] + (Bottom Arc × 0.20)

[Drive

&

Fatigue Load] + (Center Line × 0.15)[Recovery Velocity] + (Right Bar × 0.15)

[Injury Risk Premium] + (Glow Ring × 0.15) [NIL Momentum Score] + (Left Bar ×

0.10) [Substitution Probability] Score Range : 0 – 100 Target Op. Range : 60

– 85 Elite Benchmark : 78 – 88

Each raw sensor output is normalized to a 0-100 scale before weighting, ensuring consistent composite computation regardless of unit variance across hardware sources (IMU, GPS, HRV monitor). Session-level and match-level scores are stored separately and aggregated to produce a rolling 30-day BLEI-E Average used as the primary bond valuation metric.

Rugby Player Development Dashboard

The BLEI-E Dashboard is the primary visualization and monitoring interface for the index. It is designed to be deployable as a private GitHub Pages site, a Streamlit application, or an embedded web panel accessible to licensed data partners under

NDA. All data ingested into the dashboard flows through the 24-Method HEDGE capture protocol and is immutably logged before visualization.

Dashboard Components

- **Live BLEI-E Score Ticker:** Real-time composite score updated per training session and per match. Displays current score, 7-day rolling average, and trend arrow (up/neutral/down) against the 60–85 target operating range.
- **BiomechFlow Charts:** Session-over-session composite score trend lines plotted across the full training and match calendar. Includes factor-level breakdown charts showing individual zone contributions per session.
- **Heatmaps:** Contact zone frequency maps rendered on a scaled rugby pitch grid. Visualizes where on the field high-contact-force events cluster across a season — critical data for quantitative analysts and injury risk modelers.
- **Visualization Panels:** HRV recovery curves plotted per session, fatigue load progression across training blocks, and NIL momentum spike charts correlated against social engagement events and media appearances.
- **Wearable Integration:** Automated data ingestion pipelines from Catapult GPS (GPS velocity, load units, PlayerLoad), Polar H10 (HRV, heart rate, inter-beat interval), and Whoop 4.0 (recovery score, strain score, sleep performance). All streams are timestamped and merged into the BLEI-E session log at the CapsuleCycleID level.
- **AR / 3D Overlay Replay:** Visualization of 24-Method HEDGE trigger events rendered as AR/3D overlays on recorded match footage. Supports Illumix-powered spatial event tagging at 1–3 second snapshot cadence. Designed for analyst review and bond-readiness narrative documentation.

Data Capture Protocol — The 24-Method HEDGE Module

The **24-Method HEDGE Module** is the operational data engine of the BLEI-E Index. All capsule zone readings, trigger events, and composite score computations are driven by this structured 24-method framework. Every data point captured under the HEDGE protocol carries a unique CapsuleCycleID and is immutably logged to the session record before any derived computation.

Method Groupings

Method Group	Scope	Description
Methods 1-12	Core Hedge, Exit & Arbitrage Triggers	Primary capsule zone signal capture — hedge triggers, exit condition flags, and inter-zone arbitrage signals mapped to the six-factor model. Forms the foundational layer of every BLEI-E session computation.
Methods 13-20	Advanced AR / 3D Scenario Triggers	Extended spatial event capture using Illumix AR/3D overlays. Maps three-dimensional movement events — direction changes, collision vectors, positional recovery arcs — to capsule zones in real time during live sessions and matches.
Methods 21-24	Medical Escalation Protocols	Four dedicated safety and medical-grade signal methods: (21) Spatial Collapse Prediction — flags high-probability fall or collapse events; (22) Impact Vector Overload — detects contact force exceeding safe threshold bands; (23) Joint Divergence Score — monitors joint angle deviation beyond anatomical safe ranges; (24) Spatial Recovery Envelope — tracks post-contact spatial repositioning efficiency, flagging delayed recovery for medical review.

Capture Hardware Stack

- **IMU Telemetry:** Inertial measurement unit data — accelerometer, gyroscope, magnetometer — providing raw motion vectors at capsule zone level
- **Illumix AR / 3D Overlays:** Spatial event tagging and three-dimensional movement mapping integrated with live footage
- **Performance Camcorder:** 1080p / 60fps recording as the master visual reference for all trigger event verification
- **Catapult GPS + Polar H10 + Whoop 4.0:** Multi-stream wearable data ingestion (see Dashboard section)

Logging Standards

- **Snapshot Cadence:** 1–3 seconds per cycle in live game mode; 3–5 seconds per cycle in training mode
- **Log Fields:** Timestamp (UTC) | CapsuleCycleID | TriggerValue | TriggerMethod (1–24) | AnalystID | SessionType (Training/Match) | VenueID
- **Immutability:** All raw logs are append-only. No deletion or modification permitted after write. Audit trail maintained for licensing compliance.
- **Compliance Framework:** SHIELD Act (NY State biometric data protection) + NYC Local Law 3 (automated employment decisions disclosure, applied prospectively as a compliance-forward posture)

Licensing Data Agreement — Step by Step

The following is the authoritative step-by-step process Dashawn Bledsoe will execute to bring the BLEI-E dataset to market as a licensed research asset. Each step must be completed in sequence. No data shall be transmitted to any third party before Steps 1 and 2 are completed.

Step	Action	Details
STEP 1	Copyright Protection	File the BLEI-E methodology, formula, and index architecture at copyright.gov (\$65 filing fee). Establishes individual authorship, date of creation, and ownership of the 6-Factor Model and 24-Method HEDGE framework as original intellectual work.
STEP 2	NDA Execution	Any firm, institution, analyst, or individual requesting access to BLEI-E data, methodology documentation, or session logs must execute a Non-Disclosure Agreement before receiving any materials. NDA template is located in /docs/NDA-Template.pdf.
STEP 3	Initial Outreach	Send BLEI-E licensing inquiry email to target firms using the standardized outreach template. Template is located in /docs/Outreach-Email-Template.md. Do not share raw data or formula weights in the initial inquiry — share only the executive summary and index overview.
STEP 4	Data License Agreement (DLA) Negotiation	Upon NDA execution and confirmed mutual interest, negotiate a Data License Agreement. The DLA shall be non-exclusive, revocable, and time-bounded. Licensor (Bledsoe) retains full ownership of all underlying data, methodology, and derived indices at all times. DLA template in

Step	Action	Details
		/docs/Data-License-Agreement-Template.pdf.
STEP 5	Fee Structure Agreement	Negotiate and execute fee schedule: flat monthly research access fee of \$500-\$2,000 depending on data access tier, plus a 5-10% revenue share on any derived research products, published analyses, or financial instruments that incorporate BLEI-E data as an input.
STEP 6	Audit Rights	Licensor retains quarterly audit rights on all data usage by the licensee. Licensee must provide usage logs upon request within 10 business days. Unauthorized derivative products or sub-licensing without written consent triggers immediate DLA termination.
STEP 7	Term and Renewal	Initial DLA term: 12 months from execution date. Renewal is optional and requires written agreement from both parties no fewer than 30 days before expiration. Licensor may elect not to renew for any reason without penalty.
STEP 8	Governing Law	All agreements are governed by the laws of the State of New York. Any disputes arising from the DLA or related agreements shall be subject to the exclusive jurisdiction of the courts of the State of New York, County of New York.

Partnership Interest — Target Firms

The following firms represent the primary licensing and partnership outreach targets for the BLEI-E Index, ranked by fit score based on data acquisition appetite, existing rugby or athlete data infrastructure, and geographic proximity to the NYRC market.

Rank	Firm	Type	Strategic Fit
1	Sports Data Labs (SDL)	Athlete Data Rights Platform	Direct individual athlete data licensing channel. SDL is the closest structural match to the BLEI-E licensing model — purpose-built for athlete-owned data commercialization.
2	Kitman Labs	Sports Performance Data	Globally active acquirer of novel athlete performance datasets. Strong interest in underrepresented sports markets. Direct pathway to institutional-grade data packaging.
3	Stats Perform	Sports Analytics	Significant US rugby coverage gap creates a first-mover opportunity. BLEI-E provides session-level biomechanical data Stats Perform does not currently capture in the Division I club market.
4	Two Sigma Sports	Quantitative Fund	Located at 100 Ave of the Americas, NYC — five miles from NYRC. Active

Rank	Firm	Type	Strategic Fit
			in sports data partnership structures. BLEI-E as a novel, non-correlated data signal fits quant fund research mandates.
5	Zelus Analytics	Sports Science & Analytics	Actively expanding rugby coverage internationally. BLEI-E provides a domestic Division I node that Zelus does not have instrumented. Potential for multi-season data partnership.
6	Orreco	Athlete Data & Science Platform	Partnership-friendly organizational culture with demonstrated appetite for working with individual athletes. Orreco's biometric data infrastructure aligns directly with BLEI-E's HRV and fatigue load signals.

Repository Structure

The `research-blei-e` repository is organized into the following directory structure. All documents in `/legal` and `/docs` require NDA execution before access is granted to any third party. Raw session data in `/data` is never published publicly.

research-blei-e/		—	README.md		—	/docs		—	BLEI-E-Index-v1.0.pdf		—	BLEI-E-Bond-Framework-v1.0.pdf		—	NDA-Template.pdf		—	Data-
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License-Agreement-Template.pdf | └─ Outreach-Email-Template.md |──
/data | |── session-log-template.csv | └─ blei-e-score-tracker.xlsx |──
/dashboard | |── biomechflow-charts/ | |── heatmaps/ | └─ wearable-
integration/ |── /legal | |── copyright-filing-record.pdf | └─ IP-ownership-
statement.pdf └─ /whitepapers └─ Biometric-Index-Market-NY-Knicks-2026-
NBA-Finals-Case-Study.docx

Benchmarking

The following benchmarks define the BLEI-E Score performance tiers against which Dashawn Bledsoe's composite score is evaluated. Benchmarks are derived from estimated peer cohort ranges and professional rugby performance data. They will be recalibrated as licensed partner data enriches the reference pool.

Benchmark Tier	BLEI-E Score Range
Elite Professional Rugby	78 - 88
Division I Club Peer (NYRC)	58 - 72
Rookie Baseline Target	55 - 68
12-Month Growth Target (Bledsoe)	65 - 78

Bond Readiness Scorecard

The Bond Readiness Scorecard tracks the minimum criteria required before the Dashawn Bledsoe Bond can be presented to accredited investors under a formal Reg D 506(b) private placement structure. All criteria must reach Target status before any commercial bond activity commences.

Criteria	Target	Current Status
BLEI-E Dataset Sessions Logged	20+ sessions	Building
Copyright Filed (copyright.gov)	Yes	Pending — copyright.gov
NDA Template Ready	Yes	In Progress
First Data License Agreement Signed	1+ DLA executed	Outreach Phase
BLEI-E Score Benchmark	65+ composite	Training Progression
Social / NIL Presence	5,000+ followers	Building
Securities Attorney Engaged	Yes	Required before issuance
Accredited Investor Network	3+ contacts	Columbia / LinkedIn outreach

Disclaimer

⚠ IMPORTANT LEGAL NOTICE — READ BEFORE PROCEEDING

This repository and all contained documents are authored by **Dashawn Ramel Bledsoe** as a personal quantitative research methodology and intellectual property project. Nothing herein constitutes a securities offering, financial advice, investment solicitation, or an offer to buy or sell any financial instrument.

All bond framework references, coupon structures, maturity triggers, par values, and investor-facing language contained in this repository are for **research and conceptual development purposes only**. No bond has been issued. No investment is currently being accepted. No securities registration has been filed.

Any formal bond issuance requires full **SEC Regulation D 506(b) compliance**, independent legal counsel from a licensed securities attorney,

and engagement with a registered broker-dealer or placement agent where required by applicable law. Do not rely on any content in this repository as legal, financial, or investment advice.

All data licensing agreements require separate execution of binding legal documentation. Consult a licensed securities attorney and a licensed data-rights attorney before executing any commercial agreement arising from materials in this repository.

Governing Law: State of New York. All rights reserved. © 2026 Dashawn Ramel Bledsoe. Unauthorized reproduction or distribution of any portion of this methodology is prohibited.

Contact / Licensing Inquiries

Channel	Details
GitHub Repository	github.com/drb-apes/research-blei-e
Licensing Inquiries	Open an Issue on this repository tagged <code>licensing-inquiry</code> . Include your organization name, intended use case, and preferred data tier. All inquiries reviewed within 5 business days.
Partnership Interest	Open an Issue tagged <code>partnership-interest</code> or contact via the link in bio on the GitHub profile page.
Research Collaboration	Open an Issue tagged <code>research-collaboration</code> . Academic and non-commercial research inquiries from accredited institutions are welcome subject to NDA execution.
NDA Requests	Request NDA package via Issue or direct contact. NDA will be provided within 3 business days of confirmed identity and organizational affiliation.

Contributing / Pull Requests

This is a proprietary research repository. Pull requests from external contributors will not be accepted. Issues and licensing inquiries are the only open channels. All data, methodology, and codebase remain under sole authorship and ownership of Dashawn Ramel Bledsoe.

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Division I USA Rugby

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github.com/dr-b-apes/research-blei-e

This document is the authoritative public-facing project proposal for the BLEI-E Index repository. For any version other than the current GitHub release, verify authenticity via the repository commit history.